

Digital Business



Topic 7

E-Business Security

Learning Objectives

- 7.1** Understand the scope of e-business crime and security problems, the key dimensions of e-business security, and the tension between security and other values.
- 7.2** Identify the key security threats in the e-business environment.
- 7.3** Describe how technology helps secure Internet communications channels and protect networks, servers, and clients.
- 7.4** Appreciate the importance of policies, procedures, and laws in creating security.
- 7.5** Identify the major e-business payment systems in use today.

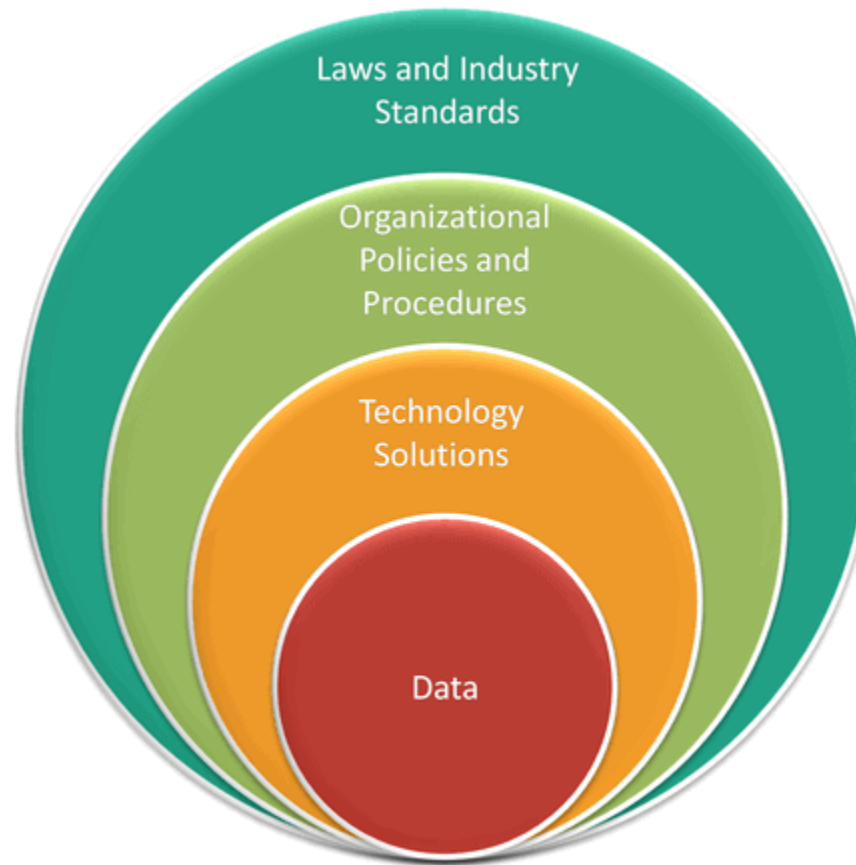
The E-business Security Environment

- Scope of the problem
 - Overall size of and losses due to cybercrime unclear
 - McAfee/Center for Strategic and International Studies study: Global economic impact of cybercrime and cyberespionage between \$455 billion to \$600 billion
 - Reports by security product providers indicate increasing cybercrime
 - Online credit card fraud one of the most high-profile forms
- Underground economy marketplaces sell stolen information, malware and more

What is Good e-business Security?

- To achieve highest degree of security
 - New technologies
 - Organizational policies and procedures
 - Industry standards and government laws
- Other factors
 - Time value of money
 - Cost of security vs potential loss
 - Security often breaks at weakest link

The e-business Security Environment



<https://www.sketchbubble.com/en/presentation-ecommerce-security.html>

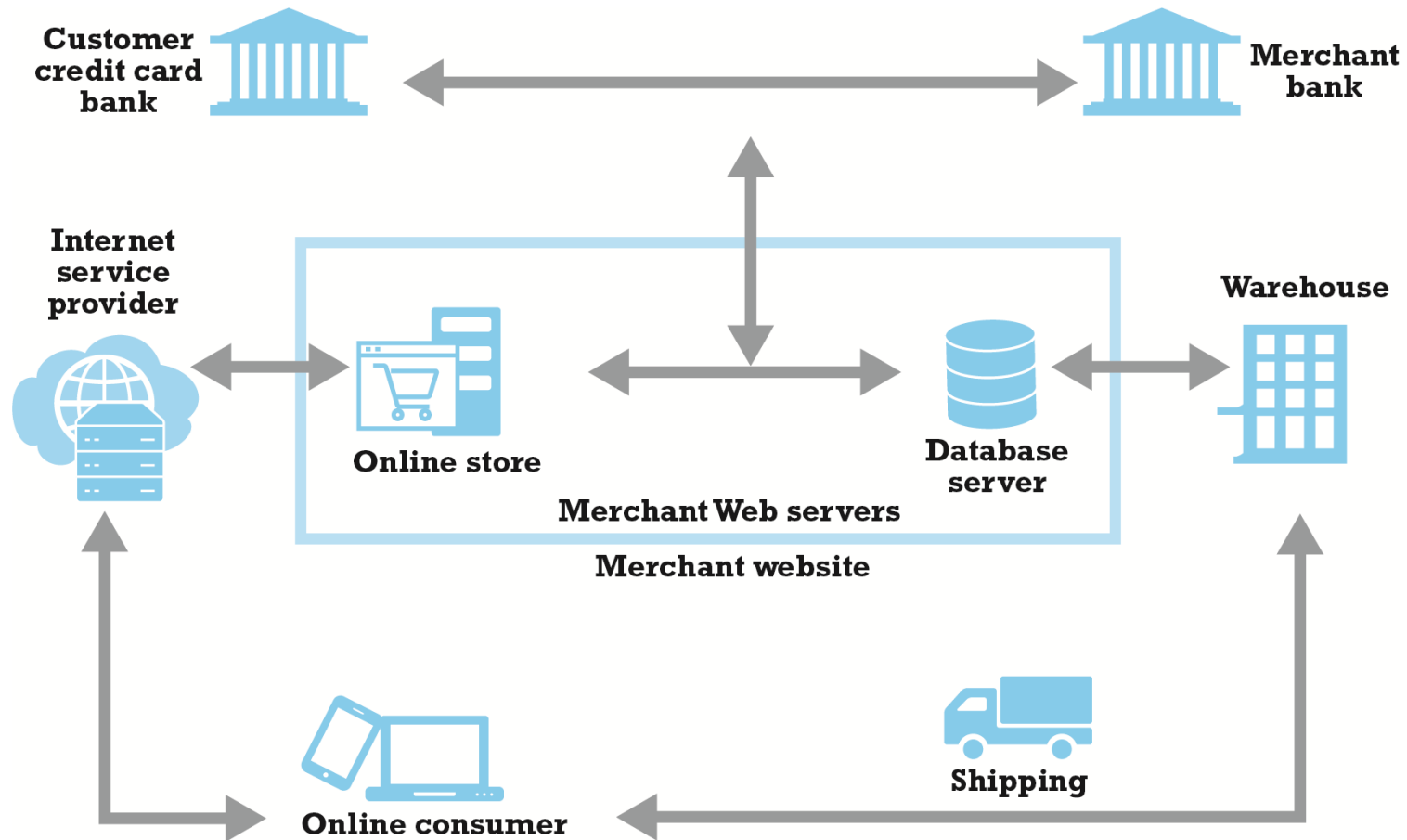
The Tension between Security and Other Values

- Ease of use
 - The more security measures added, the more difficult a site is to use, and the slower it becomes
- Public safety and criminal uses of the Internet
 - Use of technology by criminals to plan crimes or threaten nation-state

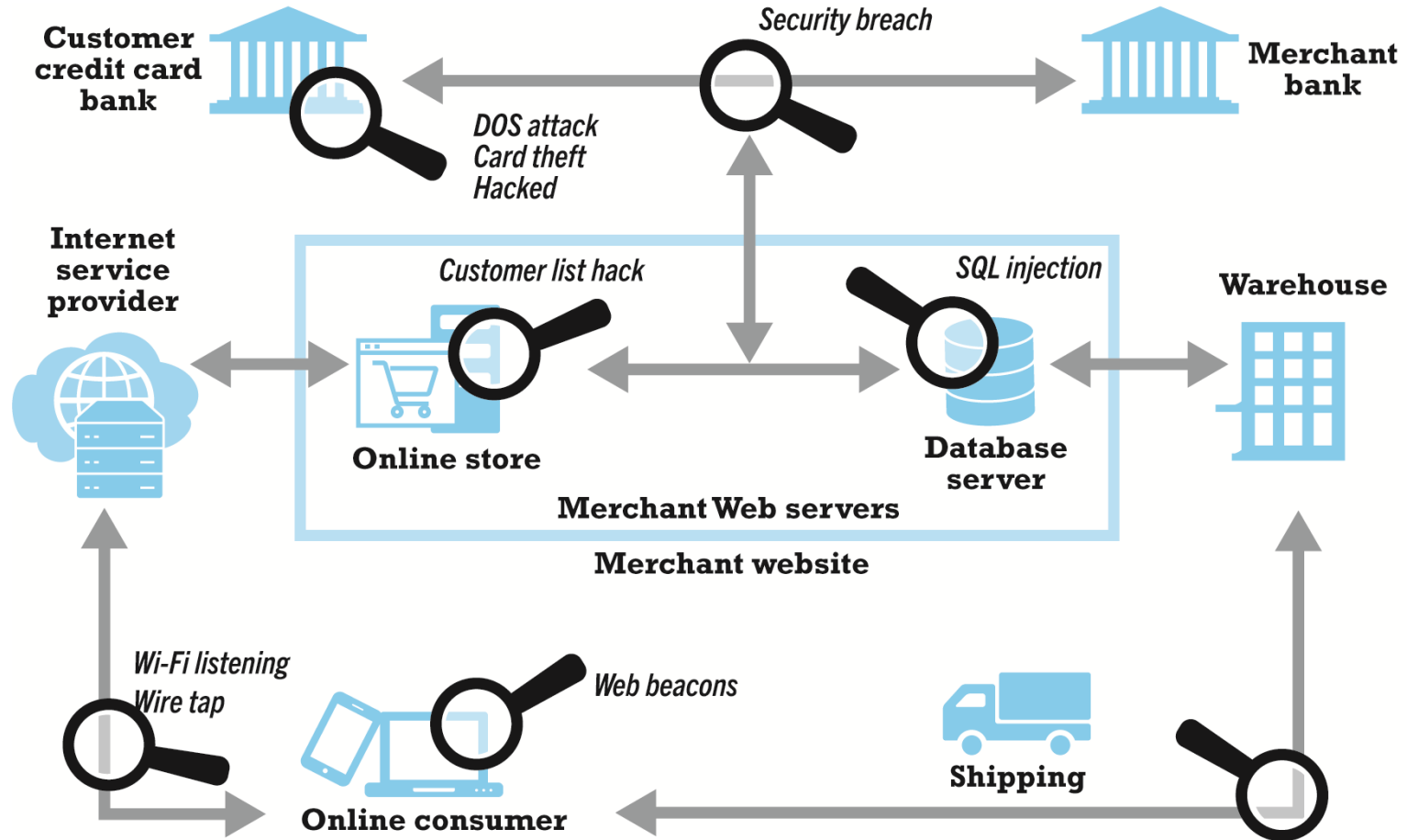
Security Threats in the e-business Environment

- Three key points of vulnerability in e-business environment:
 - Client
 - Server
 - Communications pipeline (Internet communications channels)

A Typical e-business Transaction



Vulnerable Points in an e-business Transaction

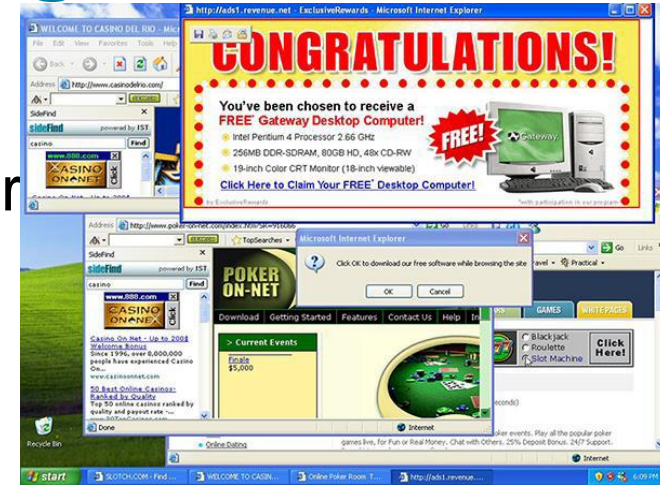


Malicious Code

- Exploits and exploit kits
- Malvertising
- Drive-by downloads
- Viruses
- Worms
- Ransomware
- Trojan horses
- Backdoors
- Bots, botnets

Potentially Unwanted Programs

- Browser parasites
 - Monitor and change user's browser
- Adware
 - Used to call pop-up ads
- Spyware
 - Tracks users keystrokes, e-mails, IMs, etc.



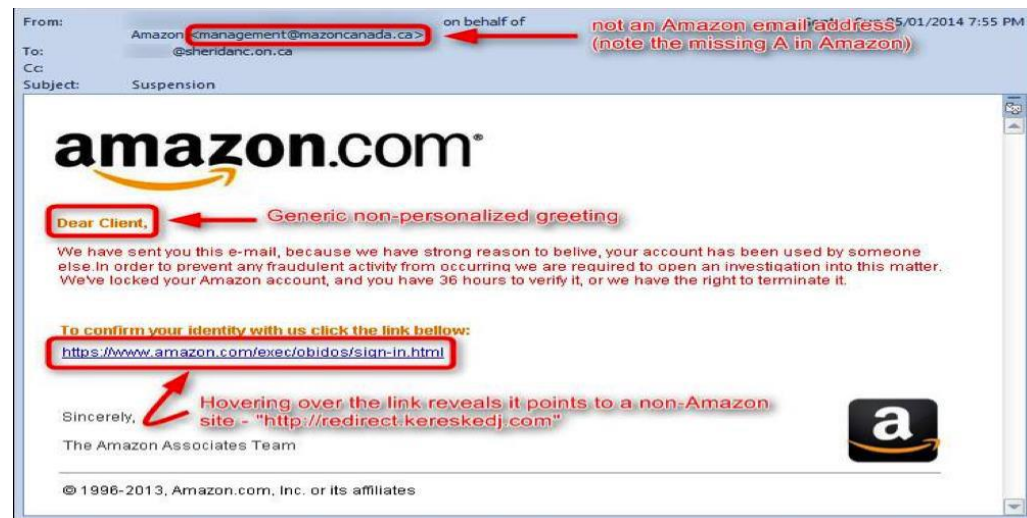
Hacking, Cybervandalism, and Hacktivism

- Hacking
 - Hackers vs crackers
 - White hats, black hats, grey hats
 - Tiger teams
 - Goals: cybervandalism, data breaches
- Cybervandalism:
 - Disrupting, defacing, destroying Web site
- Hacktivism



Phishing

- Any deceptive, online attempt by a third party to obtain confidential information for financial gain
- Tactics
 - Social engineering
 - E-mail scams and BEC phishing
 - Spear phishing
- Used for identity fraud and theft



OCBC says S\$13.7 million lost in phishing scams, up from S\$8.5 million



Vanessa Paige Chelvan

30 Jan 2022 05:57PM

(Updated: 30 Jan 2022 07:21PM)



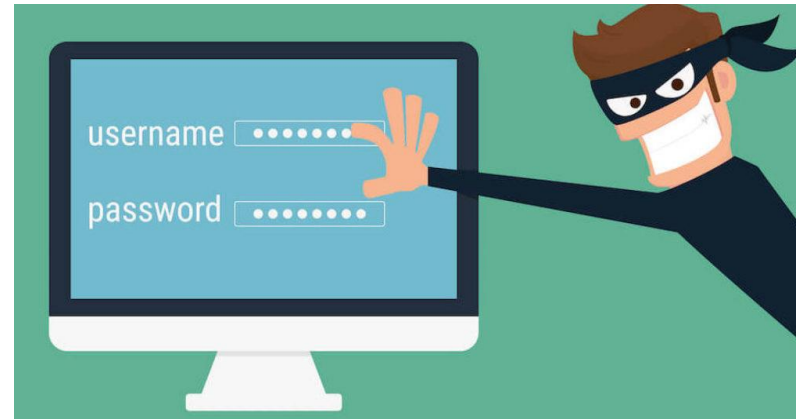
Credit Card Fraud/Theft

- Stolen credit card incidences about 0.9% on the Web and about 0.8% of mobile transactions
- Hacking and looting of corporate servers is primary cause
- Central security issue: establishing customer identity
 - E-signatures
 - Multi-factor authentication
 - Fingerprint identification



Identity Fraud/Theft

- Unauthorized use of another person's personal data for illegal financial benefit
 - Social security number
 - Driver's license
 - Credit card numbers
 - Usernames/passwords
- 2017: Almost 17 million U.S. consumers suffered identity fraud



Social Network Security Issues

- Social networks an environment for:
 - Viruses, site takeovers, identity fraud, malware-loaded apps, click hijacking, phishing, spam
- Manual sharing scams
 - Sharing of files that link to malicious sites
- Fake offerings, fake Like buttons, and fake apps

Internet of Things Security Issues

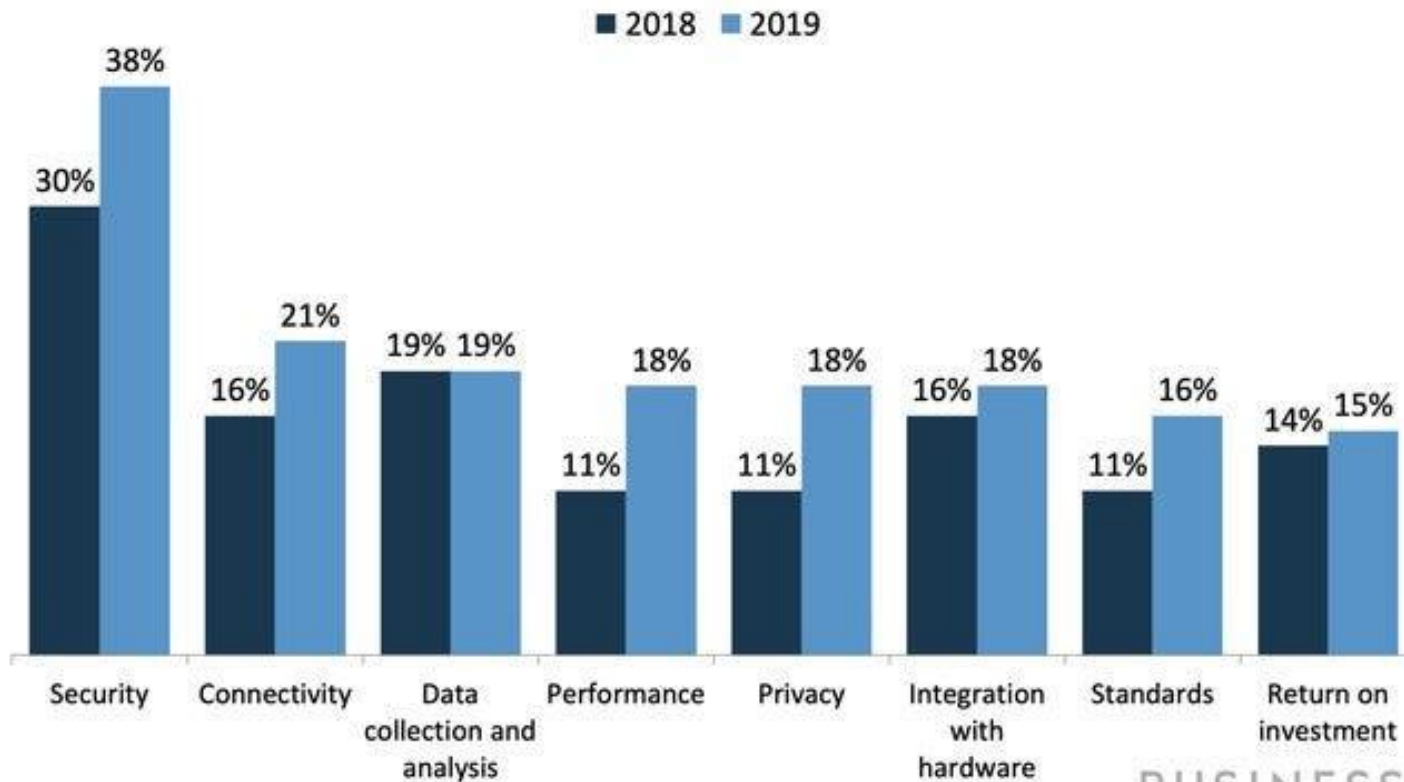
- Challenging environment to protect
- Vast quantity of interconnected links
- Near identical devices with long service lives
- Many devices have no upgrade features
- Little visibility into workings, data, or security



Internet of Things Security Issues

Security Continues To Concern IoT Developers

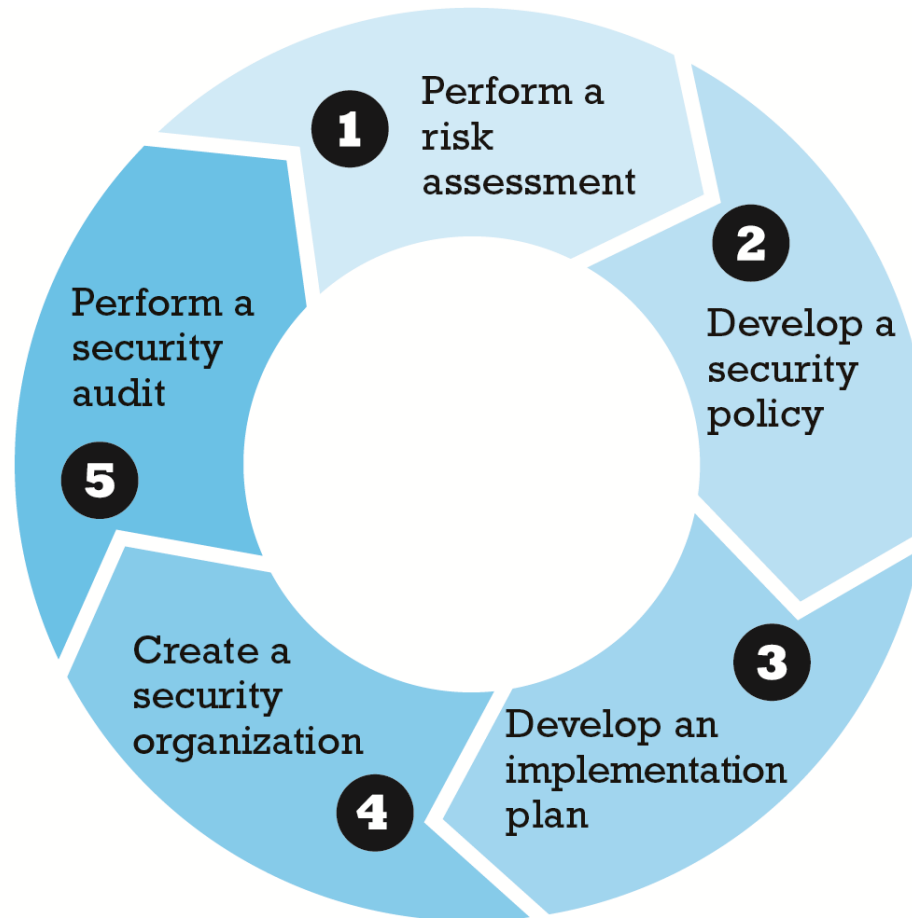
Q: What are your top two concerns for developing IoT solutions?



Source: Eclipse IoT Developer Surveys, n=502, 2018; n=1,717, 2019

BUSINESS
INSIDER
INTELLIGENCE

Developing an e-business Security Plan



Technology Solutions

- Protecting Internet communications
 - Encryption
- Securing channels of communication
 - SSL, TLS, VPNs, Wi-Fi
- Protecting networks
 - Firewalls, proxy servers, IDS, IPS
- Protecting servers and clients
 - OS security, anti-virus software



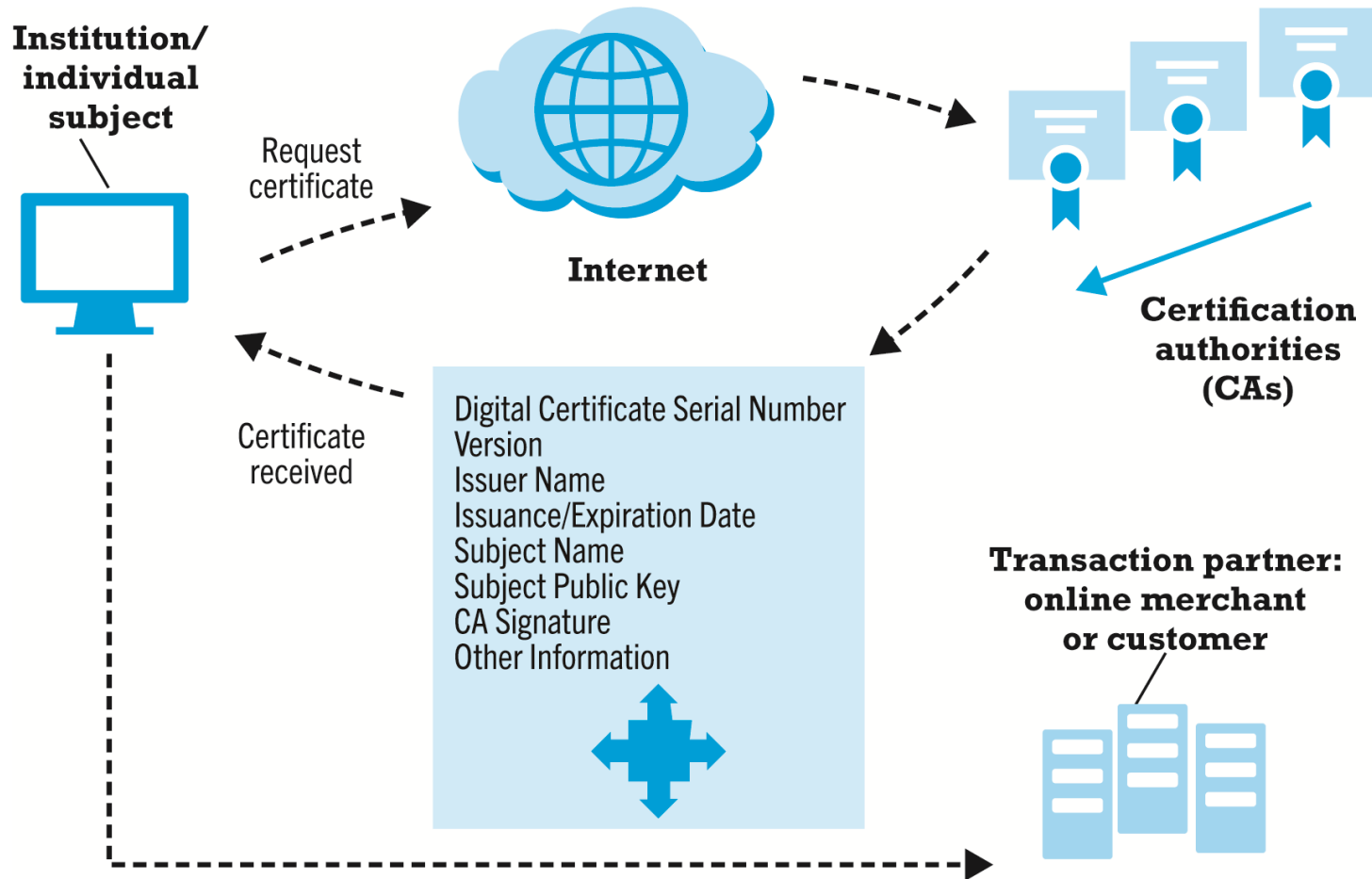
Encryption

- Encryption
 - Transforms data into cipher text readable only by sender and receiver
 - Secures stored information and information transmission
 - Provides 4 of 6 key dimensions of e-business security:
 - Message integrity
 - Nonrepudiation
 - Authentication
 - Confidentiality

Digital Certificates and Public Key Infrastructure (PKI)

- Digital certificate includes:
 - Name of subject/company
 - Subject's public key
 - Digital certificate serial number
 - Expiration date, issuance date
 - Digital signature of CA
- Public Key Infrastructure (PKI):
 - CAs and digital certificate procedures
 - PGP

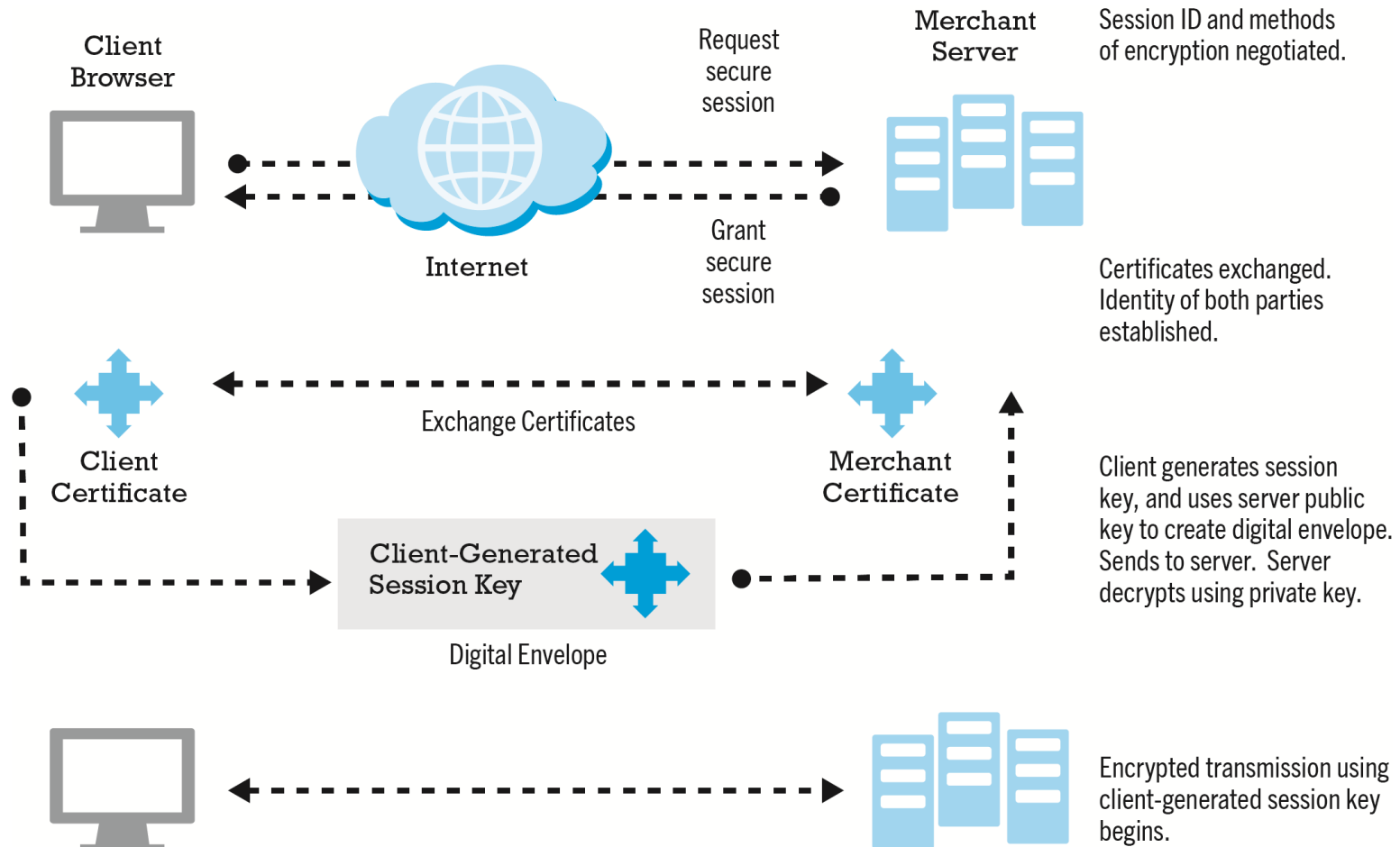
Digital Certificates and Certification Authorities



Securing Channels of Communication

- Secure Sockets Layer (SSL)/Transport Layer Security (TLS)
 - Establishes secure, negotiated client-server session
- Virtual Private Network (VPN)
 - Allows remote users to securely access internal network via the Internet
- Wireless (Wi-Fi) networks
 - WPA2
 - WPA3

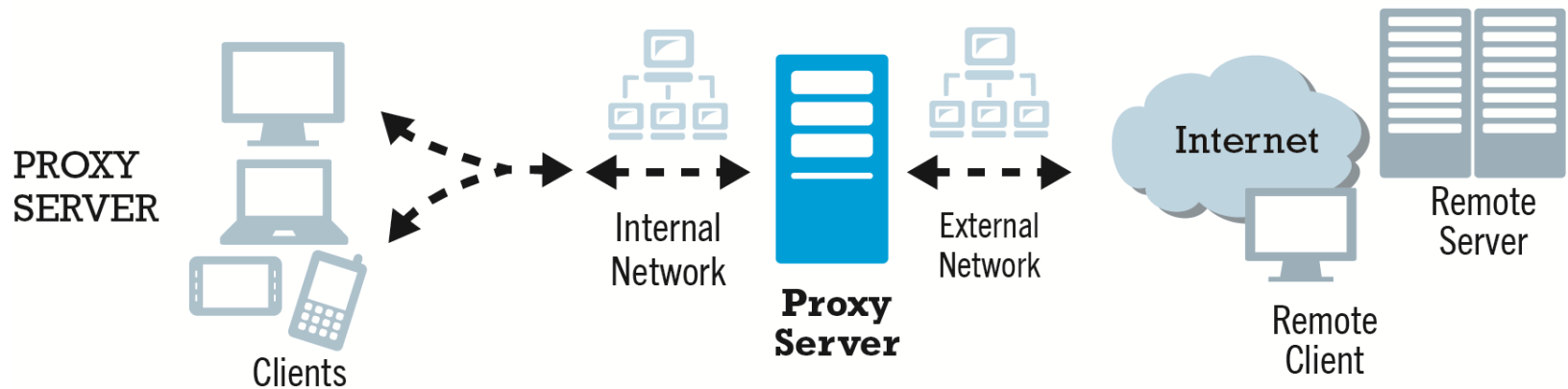
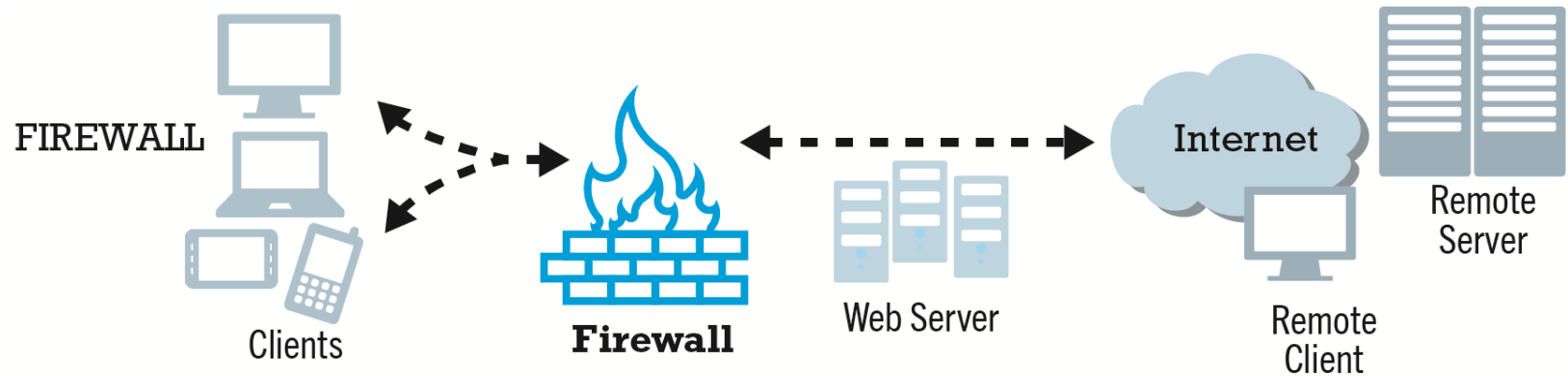
Secure Negotiated Sessions Using SSL/TLS



Protecting Networks

- Firewall
 - Hardware or software that uses security policy to filter packets
 - Packet filters
 - Application gateways
 - Next-generation firewalls
- Proxy servers (proxies)
 - Software servers that handle all communications from or sent to the Internet
- Intrusion detection systems
- Intrusion prevention systems

Firewalls and Proxy Servers



Protecting Servers and Clients

- Operating system security enhancements
 - Upgrades, patches
- Anti-virus software
 - Easiest and least expensive way to prevent threats to system integrity
 - Requires daily updates

Management Policies, Business Procedures, and Public Laws

- Worldwide, companies spend more than \$86 billion on security hardware, software, services
- Managing risk includes:
 - Technology
 - Effective management policies
 - Public laws and active enforcement

Questions??